

Beijing–Hangzhou–Zurich Moduli Workshop

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Matrices and Moduli

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Large- N expansions and geometry

In **gauge theories**, Feynman diagrams carry a natural topology.

In the large- N limit, n -point correlators admit a genus expansion

$$\sum_{g \geq 0} N^{2-2g-n} \text{ (genus } g\text{)}.$$

This suggests a hidden description in terms of a **string theory**.

Dream of gauge/string duality

Write gauge-theory correlators as integrals of natural classes on $\overline{\mathcal{M}}_{g,n}$.

Examples

- double-scaled matrix models/minimal strings
- Gopakumar–Vafa/Gromov–Witten

Applications

- Harer–Zagier
- Witten–Kontsevich

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Goal of today

Starting point: Hermitian matrix model, $U(N)$ gauge theory

Question, after 't Hooft '74

Can one construct an explicit string dual to a given Hermitian matrix model?

Goal:

$$\left\langle \prod_{i=1}^n \text{Tr} M^{k_i} \right\rangle_g = \int_{\overline{\mathcal{M}}_{g,n}} \left\langle \left\langle \prod_{i=1}^n \mathcal{V}_{k_i} \right\rangle \right\rangle_g$$

- LHS: interacting Hermitian one-matrix model
- RHS: B-twisted Landau–Ginzburg model coupled to gravity

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The Gaussian unitary model

Let M be an $N \times N$ Hermitian matrix. Define the **GUE correlators**

$$\langle f(M) \rangle^\bullet := \int_{\mathcal{H}_N} f(M) e^{-N \operatorname{Tr} \frac{M^2}{2}} dM.$$

We will study products of **single-trace observables**: $f(M) = \prod_{i=1}^n \operatorname{Tr} M^{k_i}$.

The basic Gaussian expectation value is

$$\langle M_{ij} M_{kl} \rangle^\bullet = \frac{1}{N} \delta_{ij} \delta_{kl}.$$

All higher correlators of traces are then computed by **Wick's rule**:

$$\langle M_{i_1 j_1} \cdots M_{i_{2m} j_{2m}} \rangle^\bullet = \sum_{\substack{\pi \\ \text{pairings of } 2m}} \prod_{(a,b) \in \pi} \langle M_{i_a j_a} M_{i_b j_b} \rangle^\bullet.$$

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Theorem ('t Hooft, Brézin–Itzykson–Parisi–Zuber, Harer–Zagier)

$$\left\langle \prod_{i=1}^n \text{Tr } M^{k_i} \right\rangle = \sum_{g \geq 0} N^{2-2g-n} \left\langle \prod_{i=1}^n \text{Tr } M^{k_i} \right\rangle_g$$

Notice that $\langle \prod_{i=1}^n \text{Tr } M^{k_i} \rangle_g$ is a non-negative integer, eventually zero.

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Beyond GUE: interacting matrix models

Now deform the Gaussian measure by a polynomial **potential**

$$V(M) := \sum_{k=3}^d \frac{t_k}{k} M^k, \quad S(M) := \frac{M^2}{2} - V(M).$$

Define the correlators

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We will again consider $f(M) = \prod_{i=1}^n \operatorname{Tr} M^{k_i}$.

- Expanding $e^{N \operatorname{Tr} V(M)}$ reduces everything to GUE
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$$\text{Fix } V(M) = \sum_{k=3}^d \frac{t_k}{k} M^k.$$

Matrix-model correlators are generating series of maps:

$$\left\langle \prod_{i=1}^n \text{Tr } M^{k_i} \right\rangle_g \in \mathbb{Z}[[t_3, \dots, t_d]]$$

encodes genus- g maps with prescribed boundary lengths.

For fixed g , this series is convergent for t_k sufficiently small.

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For finite N , the matrix model is a system of N interacting eigenvalues.

In the large- N limit, the discrete eigenvalues condense into a continuous **spectral density** $\rho(x)$.

This is analogous to temperature or pressure in thermodynamics: a new collective object emerges from many degrees of freedom.

Key point

Spectral density $\rho(x)$ emerges in the large- N limit

Example: GUE

$$\rho(x) = \frac{1}{2\pi} \sqrt{4 - x^2}$$



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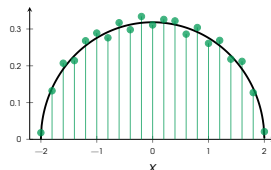
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The spectral curve

The large- N limit of the Schwinger–Dyson equations resums into an algebraic curve, called the **spectral curve**:

$$\mathcal{S} : y^2 = S'(x)^2 + P(x), \quad y(x) := \left\langle \text{Tr} \frac{1}{x - M} \right\rangle_{g=0} - \frac{S'(x)}{2}.$$

The curve is rational and can be parametrized by

$$\begin{cases} x(z) = \gamma \left(z + \frac{1}{z} \right) + \delta, \\ y(z) = \frac{1}{2} \left(z - \frac{1}{z} \right) Q(x(z)), \end{cases}$$

where γ , δ , and Q are explicitly determined from the potential V .

The spectral density is recovered from y :

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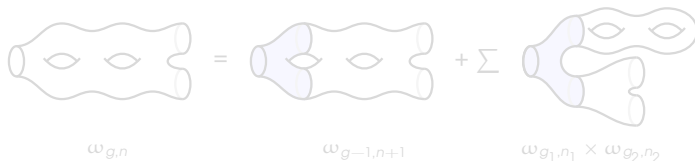
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Topological recursion: geometry $\Rightarrow$ full solution

The correlation differentials

$$\omega_{g,n}(z_1, \dots, z_n) := \left\langle \prod_{i=1}^n \text{Tr} \frac{dx(z_i)}{x(z_i) - M} \right\rangle_g$$

are determined recursively from the spectral curve.



$$\omega_{g,n} = \omega_{g-1,n+1} + \sum \omega_{g_1,n_1} \times \omega_{g_2,n_2}$$

(Chekhov–Eynard–Orantin)

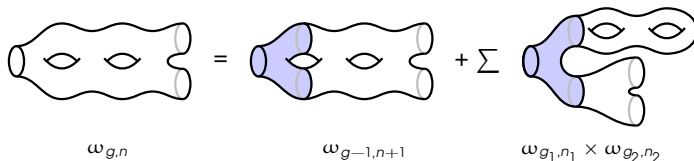
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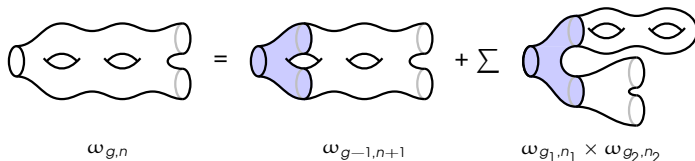
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Landau–Ginzburg w/o gravity

Inputs of a B-twisted **Landau–Ginzburg model**:

- 1d target space $X = \bar{X} - \{\text{pnts}\}$
- Holom. superpotential $W: X \rightarrow \mathbb{C}$ with non-deg. critical pnt
- Calabi–Yau form Ω

Example: B-twisted $\mathbb{CP}^1$

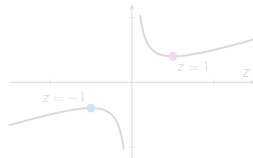
$$X = \mathbb{C}^*, \quad W(z) = z + \frac{1}{z}, \quad \Omega = \frac{dz}{z}$$

$$\mathcal{H} = \frac{\mathbb{C}[z]}{(z^2 - 1)} \cong \mathbb{C}_+ \oplus \mathbb{C}_-$$

Properties:

- Theory of maps $f: C \rightarrow X$
- Localises on **constant maps** at the critical points of W
- **State space**: Jacobi ring

$$\mathcal{H} := \frac{\mathcal{O}(X)}{(dW)} \cong \bigoplus_{dW(\alpha)=0} \mathbb{C}_\alpha$$



Critical points of W
label the states

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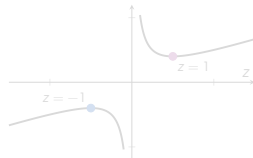
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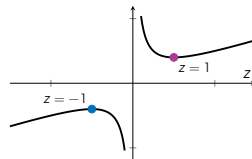
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Landau–Ginzburg w/o gravity: output

Inputs: (X, W, Ω) .

Output: a Frobenius algebra / 2d TFT on $\mathcal{H}$

$$\langle \cdot \rangle_g : \mathcal{H}^{\otimes n} \longrightarrow \mathbb{C}$$

defined via residues at the critical points of W :

$$\left\langle \prod_{i=1}^n F_i \right\rangle_g := \sum_{dW(\alpha)=0} \operatorname{Res}_{\alpha} \left(\prod_{i=1}^n F_i \right) H^g \frac{\Omega^2}{dW},$$

where $H := \frac{\nabla dW}{\Omega^2}$.

(Vafa)

The theory is **semisimple**; critical points of W label the **idempotents** $\mathcal{O}_{\alpha}$:

$$\left\langle \prod_{i=1}^n \mathcal{O}_{\alpha_i} \right\rangle_g = \delta_{\alpha_1, \dots, \alpha_n} H_{\alpha_1}^{g-1}, \quad H_{\alpha} = H(\alpha).$$

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What changes:

- the complex structure on the worldsheet C becomes dynamical
- one integrates over $\overline{\mathcal{M}}_{g,n}$

What stays:

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- the same state space $\mathcal{H}$
- symmetry and factorisation under degeneration of the worldsheet

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Output: to n states in $\mathcal{H}$, assign a cohomology class

$$\langle\langle \cdot \rangle\rangle_g : \mathcal{H}^{\otimes n} \longrightarrow H^\bullet(\overline{\mathcal{M}}_{g,n}, \mathbb{C}).$$

These classes satisfy the two basic requirements expected from a worldsheet theory:

- i) **Symmetry.** S_n -invariance.
- ii) **Gluing.** Considering the gluing maps

$$\rho: \overline{\mathcal{M}}_{g-1, n+2} \longrightarrow \overline{\mathcal{M}}_{g, n},$$

$$\sigma: \overline{\mathcal{M}}_{g_1, 1+n_1} \times \overline{\mathcal{M}}_{g_2, 1+n_2} \longrightarrow \overline{\mathcal{M}}_{g, n}, \quad g_1 + g_2 = g, \quad n_1 + n_2 = n,$$

we require

$$\rho^* \langle\langle \cdots \rangle\rangle_g = \eta^{\alpha\beta} \langle\langle \cdots \mathcal{O}_\alpha \mathcal{O}_\beta \rangle\rangle_{g-1},$$

$$\sigma^* \langle\langle \cdots \rangle\rangle_g = \eta^{\alpha\beta} \langle\langle \cdots \mathcal{O}_\alpha \rangle\rangle_{g_1} \langle\langle \cdots \mathcal{O}_\beta \rangle\rangle_{g_2}.$$

Here $\eta_{\alpha\beta} = \langle \mathcal{O}_\alpha \mathcal{O}_\beta \rangle_{g=0}$.

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Coupling LG to gravity

Output: to n states in $\mathcal{H}$, assign a cohomology class

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A worldsheet Feynman calculus

The LG + gravity path integral localizes onto **degenerations of the worldsheet** encoded by stable graphs.

Two basic building blocks appear naturally:

$$T^\alpha(u) = u - \sqrt{\frac{uH_\alpha}{2\pi}} \int_{\Gamma_\alpha} e^{-\frac{W-W_\alpha}{u}} \Omega,$$
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- T : controls the string background
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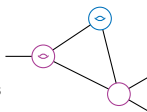
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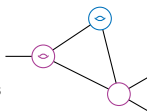
- **Graphs** $\leftrightarrow$ worldsheets built by gluing
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Resulting structure

The gravity-coupled correlators are defined from the TFT, T and R by the Givental action:

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Gauge/string duality

Theorem (Duality statement)

Every Hermitian matrix model is dual to a
B-twisted Landau–Ginzburg theory coupled to gravity

Input dictionary: fix the matrix model potential V .

- Target space: $X = \text{spectral curve} - \{\text{pnts}\}$
- Superpotential: $W = x = \gamma(z + \frac{1}{z}) + \delta \implies \mathcal{H} = \mathbb{C}_+ \oplus \mathbb{C}_-$
- Holomorphic form: $\Omega = dy$

Operator dictionary:

$$\text{Tr } M^k \longleftrightarrow \mathcal{V}_k := \left[k! \frac{\gamma u}{1 - u\psi} e^{\delta u} \sum_{\alpha=\pm 1} \mathcal{O}_\alpha \frac{l_0(2\gamma u) + \alpha l_1(2\gamma u)}{\Omega_\alpha} \right]_k$$

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$$X = \mathbb{P}^1 - \{0, \pm i, \infty\}, \quad W(z) = z + \frac{1}{z}, \quad \Omega = \frac{1}{2} \left(1 + \frac{1}{z^2}\right) dz$$

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Its integral over $\overline{\mathcal{M}}_{1,1}$ matches the Harer-Zagier computation:

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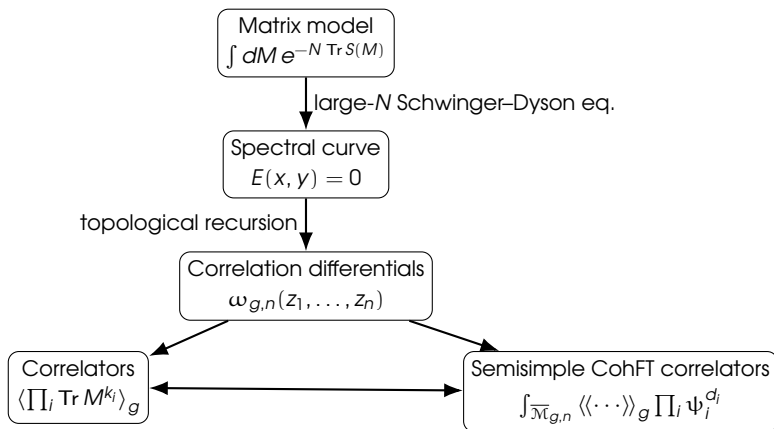
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Sketch of the proof



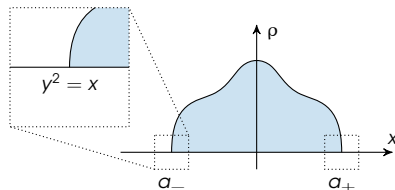
(Eynard, Dunin-Barkowski–Orantin–Shadrin–Spitz)

Edge-of-the-spectrum universality

Corollary

In the **large trace limit**:

- The spectral curve reduces to the Airy curve
- LG matter decouples from gravity



$$\frac{\langle \prod_{i=1}^n \text{Tr } M^{\kappa L_i} \rangle_g}{(2\gamma)^{\kappa|L|} \kappa^{3g-3+3n/2}} \sim \left(\frac{\langle \mathcal{O}_+^n \rangle_g}{\Omega_+^n} + (-1)^{g-1} \frac{\langle \mathcal{O}_-^n \rangle_g}{\Omega_-^n} \right)$$

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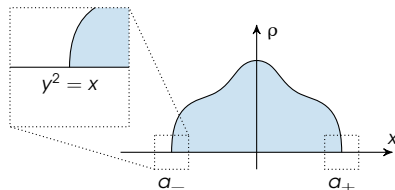
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- Topological recursion $\Leftrightarrow$ Virasoro constraints
- Topological recursion $\rightsquigarrow$ KP integrability

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DSSYK matrix model limits to WP

- **DSSYK model:** $S(M) := \sum_{k \geq 1} \frac{(-1)^{k+1}}{k} q^{\frac{k(k+1)}{2}} (1 + q^{-k}) T_{2k}(\frac{\sqrt{1-q}}{2} M)$
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Theorem (Okuyama conjecture)

For $q = e^{-\lambda}$,

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